

Calendar Anomalies: Real Patterns or Data-Mining Artifacts?

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Abstract

This paper reexamines well-known calendar anomalies in equity returns with the objective of determining whether these effects are genuine return regularities or artifacts of data mining. We study four canonical anomaly families—the day-of-the-week, week-of-the-month, month-of-the-year (January), and Sell-in-May effects—using U.S. equity data, and we also evaluate international evidence for the Sell-in-May effect. Our methodology explicitly accounts for the selection inherent in the discovery of calendar anomalies by applying data-mining-adjusted bootstrap tests that mimic a researcher’s within-family search across alternative calendar definitions. Our results show that, after correcting for data mining, the day-of-the-week, week-of-the-month, and month-of-the-year anomalies remain statistically significant in the full sample and, in particular, in the pre-1990 period, indicating that these effects were not merely statistical artifacts. At the same time, we confirm that these anomalies largely disappear in post-1990 data. In contrast, the Sell-in-May effect proves more resilient: international evidence remains robust to data-mining concerns. Overall, the findings suggest that several calendar anomalies were real features of historical return data, even though their economic relevance has diminished in more recent decades.

Keywords: calendar effects; market efficiency; data mining bias; equity returns; bootstrap methods

JEL classification: G12, G14, C12, C15

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1 Introduction

Financial economists have long used calendar time as a natural laboratory for testing market efficiency. If returns are fully unpredictable, then—aside from risk compensation—average returns should not systematically depend on whether today is a Monday rather than a Friday, the turn of the month rather than the middle, or the winter months rather than the summer. Yet a large literature reports precisely such patterns, including the day-of-the-week effect (Cross (1973), Rogalski (1984), Harris (1986)), the week-of-the-month effect (Ariel (1987), Hensel and Ziemba (1996)), the January (small-stock) effect (Rozeff and Kinney (1976), Reinganum (1983)), and the best-consecutive-six-months (Sell-in-May / Halloween-type) effect (Bouman and Jacobsen (2002), Andrade, Chhaochharia, and Fuerst (2013)). These calendar anomalies matter because they speak directly to the limits of efficiency, to models of return predictability, and to the credibility of trading rules that rely on simple and widely observable calendar labels.

A striking feature of this evidence, however, is that many calendar effects appear to be largely absent in more recent data. In particular, the day-of-the-week anomaly is reported to disappear in post-1990 samples (Kohers, Kohers, Pandey, and Kohers (2004), Olson, Mossman, and Chou (2015)). Similar conclusions have been drawn for the week-of-the-month effect (Han, Han, and Tian (2025)) and for the month-of-the-year / January-type effects (Gu (2003), Mehdiian and Perry (2002), Szakmary and Kiefer (2004)). In contrast, the Sell-in-May effect is often portrayed as comparatively more persistent. This “disappearance” pattern raises a natural question: if many calendar anomalies no longer show up out of sample, were the pre-1990 findings genuine economic regularities that were arbitrated away, or were they largely statistical artifacts that looked convincing only because of how they were discovered?

The key point is that calendar anomalies are an unusually fertile setting for data mining. The reason is structural: for any given “effect,” there are many close variants a researcher could examine—different days, weeks, months, etc.—until a striking t -statistic emerges for a calendar dummy in a linear regression. Conventional single-hypothesis p -values are not designed for this kind of search. As emphasized by White (2000), ignoring the selection step can make chance patterns look like genuine predictability. Importantly, the post-1990 disappearance of most anomalies does not reduce the relevance of this issue; rather, it makes it more pressing. Even if predictability has weakened or vanished, it remains of great interest to establish whether the

pre-1990 evidence reflected “real” patterns or was primarily an artifact of data mining.

This paper asks a focused question: among the best-known calendar effects, which ones are statistically credible once we explicitly account for the data-mining implicit in their discovery, and which ones continue to matter in more recent data? We study four canonical families in U.S. equity returns—day-of-the-week, week-of-the-month, month-of-the-year (January effect), and Sell-in-May—and we also examine whether the Sell-in-May effect is supported by cross-country evidence. The objective is to reassess prominent anomalies using tests that match how such effects are typically found and reported.

Related work most closely connected to our approach is Sullivan, Timmermann, and White (2001), who study data snooping in the context of calendar-based trading rules. They construct a very large universe of calendar strategies and apply the White Reality Check of White (2000) to assess whether the best-performing rule delivers abnormal returns relative to buy-and-hold after accounting for data mining. Their methodology is explicitly performance-based: inference is conducted on the maximum return (or Sharpe ratio) across thousands of strategies using a bootstrap procedure that corrects for the extensive search. While several calendar rules appear highly significant when evaluated in isolation, their central finding is that none remains statistically significant once the full strategy universe is taken into account.

The paper by Sullivan et al. (2001) is often interpreted as delivering a decisive verdict: calendar effects are merely artifacts of data mining. We argue that this conclusion is stronger than what their methodology can justify. The calendar-anomaly literature was not primarily concerned with whether calendar effects could be exploited to construct a profitable trading strategy, but with whether expected returns exhibit systematic seasonal patterns. Our analysis therefore revisits the question at its original econometric level by focusing on regression-based calendar effects and by explicitly accounting for the within-family specification search that characterized their discovery, rather than for optimization over a vast universe of trading rules. From this perspective, the absence of profitable calendar trading strategies does not, by itself, invalidate the statistical evidence for calendar anomalies.

Our methodological contribution is to implement data-mining-adjusted bootstrap tests that mimic a plausible researcher’s search within each anomaly family. In each bootstrap replication, we recompute the most extreme t -statistic across the relevant set of candidate dummies (e.g., across weekdays, weeks of the month, months of the year, or six-month windows), and we

compare the observed statistic to this “best-in-family” null distribution. We report results using both an i.i.d. bootstrap and a stationary block bootstrap to gauge robustness to serial dependence. For international evidence on Sell-in-May, we employ a joint bootstrap that (i) selects the best six-month window using only the bootstrapped U.S. series and then (ii) evaluates the anomaly across countries via a cross-sectional success-count statistic, thereby separating the selection step from the evaluation step.

Empirically, conventional regressions reproduce the familiar full-sample evidence for all four anomaly families. Once we apply data-mining-adjusted inference and examine subsamples, the results become more informative: the day-of-the-week, week-of-the-month, and month-of-the-year effects remain statistically credible in the full sample and, in particular, in the pre-1990 period, suggesting that these patterns were not merely artifacts of data mining. At the same time, the same procedures confirm that these three anomalies largely disappear in post-1990 data. In contrast, the Sell-in-May effect proves more robust: when evaluated using our joint bootstrap design, the international Sell-in-May evidence remains resilient to data-mining concerns.

The remainder of the paper is organized as follows. Section 2 introduces the data, the conventional regression framework, and motivating baseline evidence. Section 3 reviews related work on data snooping and calendar-based predictability. Section 4 develops our data-mining-adjusted bootstrap methodology, including the international test design for the Sell-in-May effect. Section 5 presents empirical results. Section 6 concludes.

2 Motivating Evidence and Conventional Tests

2.1 Conventional Regression Framework

A standard approach to testing for calendar anomalies is to estimate a linear regression that includes a dummy variable identifying the period associated with the anomaly of interest. Specifically, we consider the model

$$r_t = \mu + \alpha D_t + \varepsilon_t, \tag{1}$$

where r_t denotes the asset return at time t , D_t is a dummy variable equal to one when observation t falls within the anomaly window and zero otherwise, μ represents the unconditional mean return, and ε_t is an error term with zero mean.

The definition of the dummy variable depends on the anomaly under consideration. For the day-of-the-week effect, often referred to as the Monday effect, $D_t = 1$ if day t is a Monday and $D_t = 0$ otherwise. For the best consecutive six months effect (the Halloween or Sell-in-May effect), $D_t = 1$ for observations in the period from November through April and zero for the remaining months. Analogous dummy variables are constructed for the week-of-the-month and month-of-the-year effects.

The coefficient α measures the average difference in returns between periods associated with the anomaly ($D_t = 1$) and all other periods ($D_t = 0$). Under the null hypothesis of no calendar anomaly, $\alpha = 0$. A statistically significant estimate of α is therefore interpreted as evidence of abnormal return behavior linked to the corresponding calendar effect.

2.2 Data

All data used in this study, for both the US and international markets, are obtained from Kenneth French’s online data library,¹ which provides consistently constructed and widely used return series. The US data span a long period from July 1926 to December 2024, whereas the international data cover a shorter and more recent period from January 1991 to December 2024.

US data. To examine the day-of-the-week and week-of-the-month anomalies, we use daily returns on the value-weighted US stock market index. This index includes all US common stocks traded on the NYSE, AMEX, and NASDAQ and serves as a broad proxy for the US equity market.

The month-of-the-year anomaly, commonly referred to as the January effect, has been shown to be predominantly driven by small-cap stocks (see Reinganum (1983), Haug and Hirschey (2006) and Van Dijk (2011)). Accordingly, to investigate this anomaly we employ monthly returns on the Small Minus Big (SMB) factor, which captures the return differential between small- and large-cap portfolios and is constructed as the average return on three

¹http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

small-stock portfolios minus the average return on three large-stock portfolios.

To analyze the best consecutive six months effect (Sell-in-May or Halloween effect) in the US market, we use monthly returns on the value-weighted US stock market index.

International data. The international dimension of the Sell-in-May anomaly is well documented. The seminal study of Bouman and Jacobsen (2002) provides strong evidence of this anomaly across international equity markets, and subsequent work, such as Andrade et al. (2013), continues to rely on international data to assess its robustness. In line with this literature, we extend our analysis of the Sell-in-May anomaly beyond the US by examining monthly returns on value-weighted stock market indices for 20 countries.

Specifically, the countries included are: Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hong Kong, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Singapore, Spain, Sweden, Switzerland, and the United Kingdom (UK).

2.3 Baseline Evidence from Conventional Tests

Table 1 reports estimation results for the calendar anomalies using US stock market data over the period 1926–2024. Overall, the results provide strong evidence for the presence of calendar effects in the US market. In particular, the intercept estimates $\hat{\alpha}$ for the day-of-the-week, week-of-the-month, and month-of-the-year anomalies are highly statistically significant, with p -values well below 1%.

The Sell-in-May anomaly is also supported by the data. Although its p -value is slightly above the 1% level, it remains well below conventional significance thresholds, indicating strong statistical evidence in favor of this anomaly in US returns.

Table 1: Calendar Anomalies in US Stock Returns (1926–2024)

Anomaly	$\hat{\alpha}$	t -statistic	p -value
Day-of-the-Week	−0.166***	−8.827	0.000
Week-of-the-Month	0.084***	5.110	0.000
Month-of-the-Year	1.930***	5.350	0.000
Sell-in-May	0.608**	2.196	0.014

Significance levels: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Notes: This table reports estimates of several calendar anomalies based on US stock market data over the period 1926–2024. For each calendar anomaly, the intercept estimate $\hat{\alpha}$ is reported together with its t -statistic and p -value. The t -statistic is computed using Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. All reported p -values correspond to one-sided tests.

Table 2 reports estimation results for the Sell-in-May anomaly using international stock market data over the period 1991–2024. Consistent with the existing literature (see Bouman and Jacobsen (2002) and Andrade et al. (2013)), the anomaly is present in a broad cross section of international equity markets. For all countries, the estimated intercept $\hat{\alpha}$ is positive, indicating higher average returns during the November–April period than during May–October.

The statistical evidence is strong. As shown in Table 2, the Sell-in-May anomaly is significant at the 10% level in 19 out of 21 countries. At the 5% significance level, the anomaly remains significant in 16 countries, while at the more stringent 1% level it is significant in 7 countries. Overall, these results suggest that the Sell-in-May effect is not confined to a small subset of markets but instead represents a widespread international phenomenon.

Table 2: Sell-in-May Anomaly in International Stock Markets (1991–2024)

	Country	$\hat{\alpha}$	t -statistic	p -value
1	Austria	1.704***	2.796	0.005
2	Australia	1.179**	2.343	0.019
3	Belgium	1.365**	2.507	0.012
4	Canada	0.908*	1.730	0.084
5	Denmark	1.260**	2.491	0.013
6	Finland	1.514**	2.324	0.020
7	France	1.599***	3.070	0.002
8	Germany	1.373***	2.783	0.005
9	Hong Kong	0.946	1.444	0.149
10	Ireland	1.710***	3.314	0.001
11	Italy	2.079***	3.346	0.001
12	Japan	0.725	1.363	0.173
13	Netherlands	1.351**	2.426	0.015
14	New Zealand	0.975*	1.717	0.086
15	Norway	1.375**	2.327	0.020
16	Singapore	1.355**	2.131	0.033
17	Spain	1.243**	2.036	0.042
18	Sweden	1.946***	3.098	0.002
19	Switzerland	0.875***	2.577	0.010
20	UK	1.146**	2.493	0.013
21	US	0.702*	1.845	0.065

Significance levels: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Notes: This table reports estimates of the Sell-in-May anomaly for 21 international equity markets over the period 1991–2024. The intercept $\hat{\alpha}$ measures the average return differential between the November–April and May–October periods. Reported t -statistics are based on Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. All reported p -values correspond to two-sided tests.

Although the empirical results above replicate the well-known calendar anomalies docu-

mented in the literature, their correct statistical interpretation is not straightforward. The conventional t -statistics reported for the intercept estimates $\hat{\alpha}$ treat each anomaly as if it had been specified *ex ante* and therefore ignore the data-mining process underlying its discovery. In reality, these calendar effects did not arise from testing a single prespecified hypothesis, but from a sequence of closely related tests conducted within narrow families of calendar patterns.

In practice, the discovery of calendar anomalies has typically followed a targeted and incremental search strategy. Researchers examined a small set of related dummy variables—such as individual weekdays, weeks of the month, or months of the year—often estimating the same regression repeatedly while varying only the calendar indicator. An anomaly was declared whenever one of these alternatives produced a statistically significant result. Although each individual exercise involved only a limited number of tests, the overall procedure implicitly selected the most favorable outcome from a set of competing hypotheses.

When inference is conducted in this way, conventional t -statistics and p -values overstate the strength of the evidence. These statistics are valid only when the tested hypothesis is fixed independently of the data. Once significance is assessed after searching across multiple related alternatives, the probability of observing at least one apparently significant result under the null hypothesis is substantially higher than suggested by standard critical values. As a consequence, some of the detected calendar anomalies may simply reflect sampling variation amplified by within-family selection, rather than genuine return regularities.

Importantly, this concern does not rely on the existence of large-scale or exhaustive data-mining campaigns. Even modest and historically plausible search procedures—such as cycling through weekday dummies or comparing a limited number of seasonal windows—are sufficient to generate seemingly compelling findings when tests are conducted sequentially. The relevant question is therefore not whether the observed anomalies are statistically significant in isolation, but whether they remain exceptional once the selection mechanism that led to their discovery is properly taken into account.

This issue is particularly salient for the Sell-in-May anomaly, which appears to be present in a large number of international markets. While such widespread evidence may seem to provide strong support for the anomaly, international equity markets are highly correlated. As a result, similar seasonal patterns may arise simultaneously across countries even in the absence of a genuine anomaly. The appearance of the Sell-in-May effect in many markets therefore

does not constitute independent confirmation, but may instead reflect common global return dynamics. Consequently, the international evidence must also be evaluated through the lens of data mining and selection bias.

These considerations motivate the need for testing procedures that explicitly correct for within-family data mining. To determine whether the documented calendar effects reflect genuine structure in returns or merely the most favorable realizations from a small set of related hypotheses, statistical inference must incorporate the selection step inherent in their discovery. The methodology developed in this paper is designed precisely to address this issue.

3 Related Literature on Data Snooping and Calendar Effects

Sullivan et al. (2001) conducted a comprehensive study of calendar-based trading strategies and concluded that the apparent calendar effects in stock returns can largely be attributed to data mining. However, it is important to emphasize that their analysis does *not* test for the presence of specific calendar anomalies in the sense of regression (1). Instead, their objective is to determine whether an investor can systematically outperform a passive buy-and-hold strategy by exploiting any calendar-based trading rule drawn from a very large universe of such rules.

Using daily Dow Jones Industrial Average data from 1897 to 1996, they construct an extensive universe of 9,452 calendar trading rules. This rule set spans strategies based on days of the week, weeks of the month, months of the year, holidays, turn-of-the-month effects, and various combinations thereof. All standard calendar anomalies (Monday effect, January effect, first week-of-the-month effect, holiday effects, Sell-in-May effect, etc.) appear only as a tiny subset of this much broader space of possible rules.

The performance of each rule is evaluated either by its average return or by its Sharpe ratio relative to a benchmark strategy that is always fully invested in the market index. The central hypothesis is tested using the White Reality Check of White (2000), where the null states that the *best* trading rule in the entire universe does not outperform the buy-and-hold benchmark. Formally,

$$H_0 : \max_k \{E(f_k)\} \leq 0,$$

where f_k denotes the performance (excess return or excess Sharpe ratio) of rule k relative

to buy-and-hold. Conceptually, Sullivan et al. (2001) employ exactly the same methodology as Sullivan, Timmermann, and White (1999), who apply this framework to technical trading rules, but they do so in the setting of calendar-based trading rules.

To account for data mining, Sullivan, Timmermann, and White employ the stationary block bootstrap of Politis and Romano (1994) to approximate the joint distribution of performance measures across all trading rules and to obtain data-mining-adjusted p -values for the best-performing rule. Although several individual calendar rules appear significant when examined in isolation, their significance largely disappears once they are evaluated within the full universe of 9,452 competing rules. That is, after adjusting for data mining, the best calendar rule fails to deliver statistically significant outperformance relative to buy-and-hold.

The key implication for the present study is that Sullivan et al. (2001) do not test whether calendar anomalies exist in the narrow econometric sense of statistically significant dummy coefficients in regressions such as (1). Their focus is fundamentally different: they ask whether any calendar-based trading strategy, selected from a large and heavily data-mined universe, can reliably beat the market once data-snooping biases are taken into account. Thus, their conclusion that calendar effects are “the result of data mining” should be interpreted as a statement about the *profitability* and *exploitable nature* of calendar-based trading rules, rather than a statement about whether traditional calendar dummies are statistically significant.

It is precisely this latter issue—whether the statistical significance of calendar dummies reflects genuine structure in returns or instead arises from data mining within each anomaly family—that we investigate in the present study.

4 Data-Mining-Adjusted Testing Methodology

In our view, the calendar anomalies documented in the literature are likely the outcome of a series of relatively small but systematic data-mining exercises. Consider first the well-known day-of-the-week effect. In the early studies, researchers typically proceeded in a simple, incremental manner: they estimated the regression model (1) once with a dummy for Monday, checked whether the coefficient was significant, then repeated the exercise with a dummy for Tuesday, and so on for the remaining weekdays. If any one of these separate regressions produced a statistically significant coefficient—say, for Mondays—this was interpreted as evidence

of a day-of-the-week effect.

The same step-by-step logic underlies the search for other calendar patterns. To investigate month-of-the-year effects, researchers generally estimated the model twelve times, each time including a dummy for a single month and examining whether that month’s average return differed significantly from the rest. For week-of-the-month effects, the procedure was repeated using dummies for the first week, second week, and so forth. In each case, an anomaly was declared whenever one of these individually tested dummy variables produced a statistically significant result, even though significance may have arisen simply because multiple related hypotheses were explored sequentially.

A similar targeted search characterizes the discovery of the Sell-in-May (or Halloween) effect, although the search space is framed in terms of seasonal windows rather than individual calendar units. Here, researchers typically compared returns across all possible consecutive six-month intervals within the year—January–June, February–July, March–August, and so on—and identified the window that produced the largest and most significant return differential. Although only the best-performing window was ultimately highlighted, the discovery process implicitly involved several comparisons, each of which offered an opportunity for an apparently significant pattern to emerge.

Viewed in historical context, it is implausible that these early studies undertook an exhaustive search over hundreds or thousands of potential calendar partitions. Computing power in the 1980s, when most calendar anomalies were discovered, was far too limited to allow the kind of large-scale, brute-force specification search that is feasible today. Instead, researchers focused on a modest set of intuitive and easy-to-implement tests—weekday dummies, monthly dummies, weekly dummies, or simple comparisons across seasonal windows—each requiring only a small number of regressions or return comparisons. Yet even these modest, sequential searches were sufficient to produce seemingly compelling findings once several related hypotheses had been tested in turn.

Taken together, these examples illustrate that the anomalies did not arise from massive data-mining campaigns, but from repeated, targeted tests within narrow families of calendar patterns. Although each individual test was small, the cumulative effect of running many such tests increased the likelihood that at least one would appear statistically significant purely by chance, rather than revealing a genuinely robust return regularity.

In light of the discussion above, our goal is to develop a methodology for testing whether the observed calendar anomalies arise as a consequence of such data mining—that is, whether they appear in the sample purely due to luck or chance rather than reflecting genuine structure in returns. Throughout, we deliberately focus on *within-family* data mining. For each anomaly type (days of the week, weeks of the month, months of the year, and consecutive six-month windows), we examine whether the observed effect could plausibly be the outcome of pure luck when searching over the alternatives available within that family. We do not attempt to correct for, or model, possible snooping across different anomaly families. This is a deliberate design choice, consistent with the narrative that each anomaly stems from its own small search exercise.

Our objective is to test the null hypothesis that no specific calendar anomaly is present in the data. Under this null, returns are assumed to be *independent of their calendar labels*. Equivalently, the assignment of returns to calendar dates is exchangeable over time. This implies that any apparent difference between dates associated with a calendar dummy and those without one is attributable solely to sampling variation.

A crucial aspect of our bootstrap design is that, in each bootstrap replication, we reproduce the same type of search procedure that would have been used to discover the anomaly in the first place. In the empirical literature, calendar effects are typically identified by scanning across several related dummy variables and selecting the one that exhibits the strongest statistical evidence of abnormal performance. To properly account for this data-mining step, each bootstrap sample must replicate the same search procedure: we run the regression separately for each calendar dummy (e.g., Monday through Friday) and then identify the most extreme t -statistic across these individual regressions. This mirrors the way a researcher would have isolated the most significant anomaly when first examining the data. This ensures that the bootstrap distribution correctly reflects the selection mechanism inherent in the discovery of calendar effects.

Day-of-the-Week Effect

Consider the day-of-the-week anomaly. Using either the full sample or a given subsample, we estimate regression (1) including a dummy variable for Mondays. We summarize the strength

of the estimated Monday effect by the absolute value of its corresponding t -statistic,

$$t_{\text{obs}} = |t_{\text{Monday}}|.$$

Our question is then: is it plausible that a researcher, looking across the five weekdays, would end up reporting some day-of-the-week anomaly as strong as the Monday effect we observe, purely by chance under the null of no calendar pattern?

To approximate the sampling distribution of such data-mined statistics under the null, we employ bootstrap resampling of returns. Let X be the vector of daily returns of length T . In each bootstrap replication, we draw T observations from X with replacement to obtain a bootstrapped sample X^* . The calendar labels associated with each date (day, month, year) remain fixed in their original order, but the returns assigned to those dates are resampled.

In the notation r_t^* , the superscript “*” denotes the bootstrapped return at date t . Expectations under the bootstrap sampling mechanism, conditional on the original data, are denoted by $\mathbb{E}^*(\cdot)$. The conditional expectation $\mathbb{E}^*(r_t^* \mid D_t)$ therefore represents the expected value of the bootstrapped return at calendar date t , given the value of the day-of-the-week dummy.

Because returns are resampled independently of their calendar position, the bootstrap breaks any link between returns and the dummy D_t . As a result, conditional on D_t , a bootstrapped return is effectively a random draw from the empirical distribution of returns. Its conditional expectation therefore satisfies

$$\mathbb{E}^*(r_t^* \mid D_t) = \frac{1}{T} \sum_{s=1}^T r_s,$$

which is simply the unconditional sample mean. Hence, in the bootstrap world, the expected return is the same for all days of the week. This implies that the day-of-the-week dummy carries no predictive content in the resampled data, so the bootstrap provides a valid analogue of the null hypothesis of no calendar effects.

Each bootstrap replication proceeds as follows:

1. Draw T observations from the original return vector X with replacement to form the bootstrapped sample X^* .
2. Using the bootstrapped sample, estimate regression (1) for all five day-of-the-week dum-

mies and compute the associated t -statistics $t_{\text{Mon},b}^*, \dots, t_{\text{Fri},b}^*$. Define the extremal statistic

$$t_b^* = \max_{\text{day}} |t_{\text{day},b}^*|,$$

which captures the strongest day-of-the-week anomaly that a data miner could report in bootstrap replication b .

Repeating this procedure B times yields the bootstrap statistics $t_1^*, \dots, t_B^*$, which form the empirical bootstrap distribution of the data-mined test statistic under the null of no day-of-the-week effect. The bootstrap p -value is then computed as

$$p = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{t_b^* \geq t_{\text{obs}}\},$$

where B is the total number of bootstrap replications. A small p -value provides evidence against the null hypothesis of no calendar effect and suggests that the observed Monday effect is unlikely to have arisen from random variation alone, even after allowing the researcher to select the most significant weekday.

We implement this procedure using two bootstrap schemes. First, we use an i.i.d. bootstrap, which draws individual returns independently with replacement. Second, we use a stationary block bootstrap, which resamples blocks of consecutive returns to preserve the short-run dependence and volatility clustering observed in financial data. Both schemes keep calendar labels fixed and thus enforce the null that returns are independent of their calendar labels; the stationary bootstrap additionally respects the time-series dependence structure.

Week-of-the-Month and Month-of-the-Year Effects

Tests for the week-of-the-month and month-of-the-year anomalies proceed analogously. Using either the full sample or a given subsample, we estimate regression (1) including a dummy variable for the first week of each month. Similarly, we estimate regression (1) with a dummy variable for January. For these anomaly families, we take as the observed test statistics the corresponding t -statistics, denoted by $t_{\text{obs}}^{\text{W1}}$ and $t_{\text{obs}}^{\text{Jan}}$, respectively.

In each bootstrap replication, we again reproduce the original data-mining step by scanning across all relevant dummy variables and recording the most favorable t -statistic. For the week-

of-the-month effect, we estimate regression (1) with dummies for each week of the month using the bootstrapped sample and define

$$t_b^{*,\text{WOM}} = \max_{\text{week}} |t_{\text{week},b}^*|.$$

Similarly, for the month-of-the-year effect, we estimate regression (1) with twelve monthly dummies and define

$$t_b^{*,\text{MOY}} = \max_{\text{month}} |t_{\text{month},b}^*|.$$

The bootstrap p -values are then computed as the fractions of bootstrap replications in which these maximal statistics exceed the corresponding observed values, that is, the proportions for which $t_b^{*,\text{WOM}} \geq t_{\text{obs}}^{\text{W1}}$ and $t_b^{*,\text{MOY}} \geq t_{\text{obs}}^{\text{Jan}}$.

Sell-in-May Effect

To analyze the best consecutive six-month (Halloween-type) effect, we proceed analogously. Using either the full sample or a given subsample, we estimate regression (1) including a dummy variable for the November–April period and denote the corresponding test statistic (e.g., the t -statistic for the return differential) by $t_{\text{obs}}^{6\text{M}}$. In each bootstrap replication, we then compute the same statistic for all twelve possible consecutive six-month windows and record the maximum,

$$t_b^{*,6\text{M}} = \max_{\text{window}} t_{\text{window},b}^*.$$

The bootstrap p -value is subsequently obtained as the fraction of replications for which $t_b^{*,6\text{M}} \geq t_{\text{obs}}^{6\text{M}}$.

Sell-in-May Effect in International Data

Testing whether a calendar anomaly in a single return series could plausibly be the outcome of within-family data mining is conceptually straightforward, because the relevant selection step and the associated test statistic are naturally defined within that series. Extending this logic to a panel of international markets is less obvious, because the empirical evidence is inherently cross-sectional: the anomaly is judged not only by its magnitude in one country, but also by how broadly it appears across countries that are strongly comoving. To the best

of our knowledge, the literature has not provided a standard bootstrap procedure for assessing whether such *multi-country* evidence can arise purely by chance once the discovery mechanism of the Sell-in-May window is taken into account.

A key issue is therefore the choice of a cross-sectional test statistic. We summarize the international strength of the Sell-in-May evidence by counting how many countries exhibit a statistically significant return differential for the selected six-month window. Concretely, for a pre-specified significance level γ (e.g., $\gamma = 5\%$ or $\gamma = 1\%$), let $S_{\text{obs}}(\gamma)$ denote the number of countries whose t -statistic exceeds the corresponding critical value. Table 2 shows that $S_{\text{obs}}(5\%) = 16$ and $S_{\text{obs}}(1\%) = 7$. The question we address is whether an international “success count” of this magnitude could plausibly arise under the null of no seasonal return pattern, once one allows for the historically plausible data-mining step that selects the best consecutive six-month window.

To construct the appropriate null distribution, we employ a *joint* bootstrap of the international return panel that preserves contemporaneous cross-country dependence. Let X denote the $T \times 21$ matrix of monthly returns, where each row corresponds to a calendar month from January 1991 to December 2024 and each column corresponds to a country. In each bootstrap replication, we resample T rows of X with replacement to form X^* . This procedure keeps the calendar labels (month and year) fixed, while randomly reassigning returns to months. Because rows are resampled jointly across all columns, the bootstrap preserves the contemporaneous correlation structure across countries, while imposing the null that returns are independent of their calendar labels.

A crucial ingredient of the procedure is that each bootstrap replication reproduces the same discovery mechanism that historically led to the identification of the Sell-in-May anomaly. In particular, the selection of the “best” consecutive six-month window is deliberately based on U.S. returns only, reflecting the fact that the Sell-in-May effect was originally discovered in the U.S. market and subsequently examined internationally. Conditioning the window selection on the U.S. data therefore mirrors the original data-mining step underlying the anomaly. Besides, restricting the window-selection step to U.S. returns avoids “double dipping,”² as it prevents the same international data from being used both to identify the most favorable six-month

²By “double dipping” we mean the use of the same data both to select the most favorable hypothesis (here, the six-month window) and to evaluate its statistical significance, which can lead to overstated evidence against the null hypothesis.

window and to evaluate the strength of the anomaly across countries.

Each bootstrap replication proceeds as follows:

1. Draw T rows of the return matrix X with replacement to obtain the bootstrapped panel X^* .
2. Using *only* the bootstrapped U.S. return series, search across all twelve consecutive six-month windows and select the window that maximizes the t -statistic for the return differential. Denote the selected window by w_b^* .
3. Construct the dummy variable $D_{t,b}^*$ corresponding to w_b^* and estimate regression (1) separately for each country using the bootstrapped returns.
4. Compute the resulting t -statistics and record the cross-sectional success count

$$S_b^*(\gamma) = \sum_{i=1}^{21} \mathbf{1}\{t_{i,b}^* \geq c_\gamma\},$$

where c_γ is the two-sided critical value associated with significance level γ (e.g., $c_{5\%} = 1.960$ and $c_{1\%} = 2.576$ under a standard normal approximation).

Repeating this procedure for B bootstrap replications yields $S_1^*(\gamma), \dots, S_B^*(\gamma)$, which form the empirical bootstrap distribution of the international evidence under the null hypothesis of no Sell-in-May effect, while explicitly accounting for the historical U.S.-based window-selection step. The corresponding bootstrap p -value is computed as

$$p(\gamma) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{S_b^*(\gamma) \geq S_{\text{obs}}(\gamma)\}.$$

A small value of $p(\gamma)$ indicates that observing an anomaly in as many countries as reported in Table 2 is unlikely to occur by chance once one accounts for (i) the endogenous selection of the six-month window based on U.S. data and (ii) the strong cross-country dependence in returns preserved by the joint bootstrap.

5 Empirical Results with Data-Mining Corrections

5.1 US Evidence After Data-Mining Corrections

The US data span a long sample from 1926 to 2024, which allows us to examine both the overall presence of calendar anomalies and their stability over time. To assess whether statistical significance can be explained by within-family data mining, we apply the bootstrap methodology to the full sample and to three non-overlapping subsamples corresponding to the first, second, and last thirds of the sample period. The subsample analysis serves two purposes: it provides a simple robustness check and, more importantly, it reveals whether the anomalies persist in more recent data.

Table 3 reports results for the day-of-the-week anomaly (the Monday effect) across the different sample periods. For each period, we report the intercept estimate $\hat{\alpha}$ and its t -statistic. The table also reports three p -values: the nominal p -value from the conventional t -test, and data-mining-adjusted p -values obtained from an i.i.d. bootstrap and a stationary block bootstrap.³ Since the bootstrap tests are conducted as one-sided tests, the nominal p -values are also reported for one-sided alternatives.

The results provide strong evidence of a Monday effect in the full sample and in the first two subsamples. In these periods, the bootstrap p -values are essentially zero, indicating that the observed Monday effect is extremely unlikely to arise purely by chance even after allowing for the within-family search over weekdays. In other words, for the early and mid-sample periods, the Monday anomaly cannot be attributed to data mining within the day-of-the-week family.

The picture changes markedly in the most recent subsample (1994–2024). In this period, the estimated Monday coefficient is much smaller than in the other periods and the nominal p -value is about 11%, which by itself suggests at most weak evidence of a Monday effect. More importantly, once the data-mining correction is applied, the bootstrap p -values increase to roughly 66–67%. This implies that in about two thirds of the bootstrap samples generated under the null of no calendar pattern, the strongest weekday anomaly is at least as large as the

³In all bootstrap tests in this paper, inference is based on $B = 10,000$ bootstrap replications. For the stationary block bootstrap, the block length is chosen according to the rule of thumb $T^{1/3}$. We also conducted robustness checks using alternative block lengths (not reported to conserve space). The resulting p -values are very similar across specifications, indicating that the bootstrap inference is not sensitive to the precise choice of block length.

one observed in the actual data. Hence, the bootstrap tests provide no evidence of a Monday effect in the post-1994 period and suggest that the historical day-of-the-week anomaly is not robust in the most recent decades.

Table 3: Day-of-the-Week Anomaly (Monday Effect): US Evidence with Data-Mining Corrections

	Period	$\hat{\alpha}$	t -statistic	p -value		
				Nominal	i.i.d. bootstrap	Block bootstrap
1	1926–2024	-0.166	-8.827	0.000	0.000	0.000
2	1926–1959	-0.269	-7.643	0.000	0.000	0.000
3	1960–1993	-0.174	-6.364	0.000	0.000	0.000
4	1994–2024	-0.039	-1.221	0.111	0.669	0.661

Notes: This table reports tests of the day-of-the-week anomaly (Monday effect) in US daily returns for the full sample and three non-overlapping subsamples. For each period, $\hat{\alpha}$ is the intercept estimate in regression (1) with $D_t = 1$ on Mondays and $D_t = 0$ otherwise. The reported t -statistics are based on Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. “Nominal” denotes the conventional one-sided p -value based on the HAC t -statistic. “i.i.d. bootstrap” and “Block bootstrap” report data-mining-adjusted one-sided p -values obtained by (i) resampling individual returns independently and (ii) resampling blocks of consecutive returns using a stationary bootstrap, respectively, where each bootstrap replication scans across all weekdays and records the maximal absolute t -statistic within the day-of-the-week family.

Table 4 reports results for the week-of-the-month anomaly, defined as unusually high returns during the first week of each month, across the full US sample and three non-overlapping subsamples. The table shows that the conventional regression delivers statistically significant evidence of a first-week effect in the full sample and in each of the first two subsamples. However, once we account for within-family data mining by allowing a researcher to scan across the different weeks of the month and select the most favorable dummy, the strength of the evidence becomes markedly weaker in the later subsamples.

In the full sample (1926–2024) and in the early subsample (1926–1959), the anomaly is highly significant both nominally and after data-mining correction: the bootstrap p -values are essentially zero, indicating that a first-week effect of this magnitude is extremely unlikely to arise by chance under the null of no week-of-the-month pattern. In the middle subsample (1960–1993), the nominal p -value is small (0.6%), but the data-mining-adjusted p -values rise to about 5%, placing the evidence at the margin of conventional significance levels once the within-family selection step is taken into account. Finally, in the most recent subsample (1994–2024), the nominal p -value is 6.8%, whereas the bootstrap p -values increase to roughly 39–41%. Thus, in the post-1994 period, the estimated first-week effect is not unusual relative to what

would be expected under the null when a data miner is allowed to search across the weeks of the month. Overall, the bootstrap results suggest that the week-of-the-month anomaly is concentrated in the earlier part of the sample and is not robust in the most recent decades.

Table 4: Week-of-the-Month Anomaly (First-Week Effect): US Evidence with Data-Mining Corrections

	Period	$\hat{\alpha}$	t -statistic	p -value		
				Nominal	i.i.d. bootstrap	Block bootstrap
1	1926–2024	0.084	5.110	0.000	0.000	0.000
2	1926–1959	0.144	4.595	0.000	0.000	0.000
3	1960 - 1993	0.058	2.496	0.006	0.050	0.048
4	1994 - 2024	0.046	1.492	0.068	0.406	0.390

Notes: This table reports tests of the week-of-the-month anomaly (first-week effect) in US daily returns for the full sample and three non-overlapping subsamples. For each period, $\hat{\alpha}$ is the intercept estimate in regression (1) with $D_t = 1$ in the first week of the month and $D_t = 0$ otherwise. The reported t -statistics are based on Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. “Nominal” denotes the conventional one-sided p -value based on the HAC t -statistic. “i.i.d. bootstrap” and “Block bootstrap” report data-mining-adjusted one-sided p -values obtained by (i) resampling individual returns independently and (ii) resampling blocks of consecutive returns using a stationary bootstrap, respectively, where each bootstrap replication scans across all weeks-of-the-month dummies and records the maximal absolute t -statistic within the week-of-the-month family.

Table 5 reports results for the month-of-the-year anomaly (the January effect) across the full sample and three non-overlapping subsamples. The January effect is strongly present in the full sample and in the first two subsamples: both the nominal and data-mining-adjusted bootstrap p -values are essentially zero, indicating that the effect is highly unlikely to arise by chance even after allowing for within-family selection across months. In contrast, the effect disappears completely in the most recent subsample (1994–2024). In that period, the estimate of $\hat{\alpha}$ is close to zero, the nominal p -value is large, and the bootstrap p -values equal one, implying that the observed statistic is entirely unexceptional relative to the null distribution. Overall, the bootstrap evidence suggests that the January effect is concentrated in the earlier part of the sample and is not robust in the post-1994 period.

Table 6 reports results for the Sell-in-May (Halloween-type) anomaly in US monthly stock market returns across the full sample and three non-overlapping subsamples. In contrast to the other anomaly families, the evidence for Sell-in-May is much less stable over time. While the conventional regression suggests statistically significant return seasonality in the full sample and in the later subsamples, the data-mining-adjusted bootstrap tests paint a more nuanced picture: the anomaly appears to be driven primarily by the middle third of the sample.

Table 5: Month-of-the-Year Anomaly (January Effect): US Evidence with Data-Mining Corrections

	Period	$\hat{\alpha}$	t -statistic	p -value		
				Nominal	i.i.d. bootstrap	Block bootstrap
1	1926–2024	1.930	5.350	0.000	0.000	0.000
2	1926–1959	2.804	5.009	0.000	0.000	0.000
3	1960–1993	2.718	4.552	0.000	0.001	0.000
4	1994–2024	0.134	0.222	0.412	1.000	1.000

Notes: This table reports tests of the month-of-the-year anomaly (January effect) using monthly returns on the SMB factor for the full sample and three non-overlapping subsamples. For each period, $\hat{\alpha}$ is the intercept estimate in regression (1) with $D_t = 1$ in January and $D_t = 0$ otherwise. The reported t -statistics are based on Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. “Nominal” denotes the conventional one-sided p -value based on the HAC t -statistic. “i.i.d. bootstrap” and “Block bootstrap” report data-mining-adjusted one-sided p -values obtained by (i) resampling returns independently and (ii) resampling blocks of consecutive returns using a stationary bootstrap, respectively, where each bootstrap replication scans across all month dummies and records the maximal absolute t -statistic within the month-of-the-year family.

In the full sample (1926–2024), the nominal p -value is 1.4%, which would typically be interpreted as strong evidence for the anomaly. However, once we account for the discovery mechanism by scanning across all twelve consecutive six-month windows, the bootstrap p -values increase to about 11–13%. Hence, after correcting for within-family data mining, the full-sample evidence is no longer significant at conventional 10% level. This illustrates that conventional t -tests can exaggerate the strength of the evidence for the Sell-in-May effect when the best-performing window is selected endogenously.

The subsample results confirm that the anomaly is concentrated in the middle part of the sample. In the first subsample (1926–1959), the estimated return differential is essentially zero, and both bootstrap p -values equal one, indicating no evidence of seasonality. In the second subsample (1960–1990), the anomaly becomes economically larger and statistically stronger: the nominal p -value is 0.4%, while the data-mining-adjusted bootstrap p -values are around 5%. Thus, even after allowing for window selection, the Sell-in-May effect remains borderline significant in this period. In the most recent subsample (1991–2024), the nominal p -value again suggests significance (3.3%), but the bootstrap p -values rise to roughly 26%, implying that the observed statistic is not unusual once the data-mining step is incorporated. Overall, the bootstrap evidence indicates that the Sell-in-May anomaly is not a persistent feature of US returns, but is largely driven by the middle third of the sample.

Table 6: Sell-in-May Anomaly (Best Six-Month Window): US Evidence with Data-Mining Corrections

	Period	$\hat{\alpha}$	t -statistic	p -value		
				Nominal	i.i.d. bootstrap	Block bootstrap
1	1926 - 2024	0.608	2.196	0.014	0.127	0.113
2	1926 - 1959	0.046	0.077	0.469	1.000	1.000
3	1960 - 1990	1.112	2.663	0.004	0.052	0.050
4	1991 - 2024	0.702	1.845	0.033	0.278	0.248

Notes: This table reports tests of the Sell-in-May (Halloween-type) anomaly in US monthly stock market returns for the full sample and three non-overlapping subsamples. For each period, $\hat{\alpha}$ is the intercept estimate in regression (1), where $D_t = 1$ for the November–April period and $D_t = 0$ otherwise; $\hat{\alpha}$ therefore measures the average return differential between November–April and May–October. The reported t -statistics are based on Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. “Nominal” denotes the conventional one-sided p -value based on the HAC t -statistic. “i.i.d. bootstrap” and “Block bootstrap” report data-mining-adjusted one-sided p -values obtained by (i) resampling returns independently and (ii) resampling blocks of consecutive returns using a stationary bootstrap, respectively, where each bootstrap replication scans across all twelve consecutive six-month windows and records the maximal test statistic within the Sell-in-May family.

5.2 International Sell-in-May Evidence After Data-Mining Corrections

Table 7 reports bootstrap tests of the Sell-in-May anomaly using international data. As described in the methodology section, the international evidence is summarized by the cross-sectional “success count” $S_{\text{obs}}(\gamma)$, defined as the number of countries for which the estimated Sell-in-May return differential is statistically significant at a pre-specified level γ . In the original sample, the anomaly is significant in 16 out of 21 countries at the 5% level and in 7 out of 21 countries at the 1% level.

The bootstrap p -values in Table 7 adjust for two key features of the problem: (i) the endogenous selection of the best consecutive six-month window based on U.S. data, and (ii) the strong contemporaneous dependence across international markets preserved by the joint bootstrap of the return panel. The results indicate that the observed breadth of the anomaly across countries is unlikely to arise purely by chance under the null of no seasonal return pattern. Specifically, the i.i.d. and block bootstrap p -values lie between 1.8% and 2.7% across the two critical values. Hence, even after correcting for the historical data-mining step and cross-country dependence, the international manifestation of the Sell-in-May effect remains statistically significant at the 5% level. In other words, the anomaly appears weak and unstable in the U.S. time series when viewed through the lens of data mining, but it remains comparatively robust when evaluated using cross-country evidence.

Table 7: Sell-in-May Anomaly: International Evidence with Data-Mining Corrections

Critical value c_γ	Significant countries $S_{\text{obs}}(\gamma)$	Bootstrap p -value	
		i.i.d. bootstrap	Block bootstrap
1.960	16	0.018	0.021
2.576	7	0.027	0.025

Notes: This table reports bootstrap tests of the Sell-in-May anomaly using monthly returns for 21 international equity markets over the period 1991–2024. The test statistic is the cross-sectional success count $S_{\text{obs}}(\gamma)$, defined as the number of countries whose (two-sided) HAC t -statistic for the Sell-in-May return differential exceeds the critical value c_γ . In each bootstrap replication, we resample months jointly across countries to preserve contemporaneous cross-country dependence, select the best consecutive six-month window *using the bootstrapped U.S. return series only*, and then compute the success count across countries for the selected window. The reported i.i.d. and block bootstrap p -values are the fractions of bootstrap replications in which the bootstrapped success count is at least as large as the observed count.

6 Conclusion

This paper revisits some of the most prominent calendar anomalies in equity returns with a specific objective: to determine whether these effects should be interpreted as genuine return regularities or as artifacts of data mining. While much of the existing literature emphasizes either the original discovery of calendar effects or their apparent disappearance in more recent data, we focus on a complementary question—whether the historical evidence itself remains statistically credible once one explicitly accounts for the selection inherent in how such anomalies are identified.

To address this issue, we develop and implement data-mining-adjusted bootstrap tests that mirror a plausible researcher’s search within each anomaly family. Rather than testing the profitability of a single best-performing trading rule, our approach evaluates the statistical significance of classical calendar anomalies in their standard econometric form, while correcting for within-family selection. This framework allows us to separate concerns about data snooping from questions about economic persistence.

Our results lead to a clear conclusion. After accounting for data mining, the day-of-the-week, week-of-the-month, and month-of-the-year anomalies remain statistically significant in the full sample and, in particular, in the pre-1990 period. This finding suggests that these patterns were not merely statistical flukes generated by extensive searching, but reflected genuine regularities in earlier return data. At the same time, our subsample analysis confirms that these anomalies largely disappear in the post-1990 period, consistent with prior evidence

in the literature. In contrast, the Sell-in-May effect proves more resilient: when evaluated using a joint bootstrap design, the international evidence remains robust to data-mining concerns.

Taken together, our findings imply that the widespread post-1990 attenuation of calendar anomalies should not be interpreted as evidence that earlier results were spurious. Instead, the evidence is more consistent with the view that several calendar effects were real but have weakened or vanished over time, possibly due to changes in market structure, trading behavior, or arbitrage activity. More broadly, the paper illustrates the importance of aligning statistical inference with the way empirical patterns are discovered, particularly in settings—such as calendar-based predictability—where data mining is a first-order concern.

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